

Equations with Variables on Both Sides (Homework)

① $5 + 2x = 9x$ $x = 5/7$

② $-6y + 4 = 1 + 5y$ $y = 3/11$

③ $9 - 5z - 3 = -8z + 7$ $z = 1/3$

④ $\frac{2x}{3} - 1 = 1 + \frac{3x}{4}$ $x = -24$

⑤ $2x - 5 + 7x = 8x + 6$ $x = 11$

⑥ $9y + 1 = -4y + 7 + y$ $y = 1/2$

⑦ $-7z + 2 - 5z = 2z$ $z = 1/7$

⑧ $3 - 2x + x = 4x - 6x$ $x = -3$

⑨ $\frac{4x}{5} + \frac{3x}{5} = -10 + \frac{7x}{10}$ $x = -20$

⑩ $7y + 1 = 3y - 5$ $y = -3/2$

⑪ $-3 + 4z = 10z - 3z + 1$ $z = -4/3$

⑫ $-x + 8 + 5x - 6 = 2 - 4x$ $x = 0$

⑬ $-2y - 5y - 1 = 5y + 2y + 2$ $y = -3/14$

⑭ $9 + y + (-3y) = 2y$ $y = 9/4$

⑮ $z - 4 - 7z + 11 = -3 + z$ $z = 10/7$

⑯ $-8x + 2x + 1 = 7x + 3 - 2$ $x = 0$

⑰ $-7 + 4 - 2y - 9y = 7 - y$ $y = -1$

⑱ $10z + 3z + 1 = -9z + 8$ $z = 7/22$

⑲ $-x - x - 2x = -1 - 4 - 5 - 2x$ $x = 5$

⑳ $-\frac{x}{2} + \frac{3x}{4} + 1 = -\frac{x}{4} - \frac{x}{4}$ $x = -4/3$

㉑ $5y + 1 = 2y - 2$ $y = -1$

㉒ $7x - 8x + 10x = 4x - 3x + 7$ $x = 7/8$

㉓ $-20z + 3 - 9 - 4z = 8 - 2z$ $z = -7/11$

㉔ $-2 - 4 - y - 9y = -3y - 2y + 10$ $y = -16/5$

㉕ $-\frac{3x}{8} - \frac{5x}{8} + 1 = 6 - \frac{3x}{4}$ $x = -20$